# SUMMARY

**Power BI Assignment 1 – Data Transformation and Data Modeling**

**1. Data Import**

I imported three CSV files into Power BI: **List of Orders, Order Details, and Sales Target**.

**2. Data Transformation**

I used Power Query in Power BI for data transformation. I restricted the List of Orders table to the first 500 rows. I changed the Order Date column to the Date data type and converted the Amount, Target, and Profit columns to the Fixed Decimal Number data type. I also transformed the Customer Name column into proper case format.

**3. Custom Columns**

I created a Location column by combining City and State. I also created a Profit Margin column by dividing Profit by Amount and a Profit Status conditional column to classify records as Loss, Break-Even, or Profit.

**4. Data Merging**

I merged the List of Orders and Order Details tables using the Order ID column with a Left Outer join. I expanded the required columns to create a new table called Orders Data.

**5. Data Quality Checks**

I checked the data for duplicates and reviewed column quality to identify data quality issues.

**6. Sorting and Filtering**

I sorted the Orders Data table by Order Date in descending order and applied a filter to view records from Tamil Nadu.

**7. Data Summarization**

I created two summary tables:

- **Order Details Summary:** Grouped data by Category and calculated Average Profit.
- **Sales Target Summary:** Grouped data by Month of Order Date and calculated Total Target.

**8. Data Modeling**

I established the required relationships between the tables:

- List of Orders and Order Details using the Order ID column.
- Order Details and Sales Target using the Category column.

I checked that both required relationships were active in Power BI.